Physical AI — the path to a beautiful tomorrow.

PRAVDA REPLIKANTA

THE REPLICANT TRUTH
A NEWSPAPER ABOUT PHYSICAL ARTIFICIAL INTELLIGENCE

BREUKELEN (Netherlands)
Forecast · 2026-10-31 · CET

≋	MORNING · 09:00 **6.6°**	Fog Wind: E 2.6 m/s · 0%
☁	MIDDAY · 12:00 **8.1°**	Cloudy Wind: SE 3.2 m/s · 0%
☁	EVENING · 18:00 **8.9°**	Cloudy Wind: E 2.6 m/s · 0%

Daily: 5.8–11.1 °C
Precipitation probability: 0%

All we can do is learn to be human while the rain falls.

ISSUE № 3 · 2026-10-31 rev. 2 ·

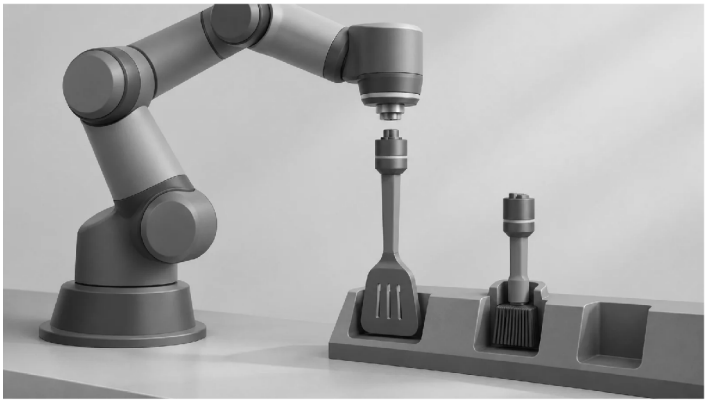
Weather source: Open-Meteo · Retrieved: 2026-10-30 19:37 CET

Replicant Pravda — Physical AI, 31 October 2026

Eight developments in embodied AI: tactile tools, simulation, recoverable task execution, Japanese warehouse deployment and the economics of scaling.

News

SwapBot replaces the universal gripper with swappable tactile tools



AI-generated illustration · Conceptual AI-generated illustration of the general SwapBot principle: a robot aligns its coupling with a tool still supported by its holder, with another tool docked nearby. Not a reconstruction of the research apparatus.

The search for a single hand that can do everything has an obvious rival: change the tool instead. SwapBot puts a standard coupling on the robot and inexpensive adapters on ordinary utensils. A spatula, brush or articulated tool can become a robot end-effector, while integrated tactile sensing reports contact that cameras alone cannot reliably reveal.

The team collected demonstrations through teleoperation and trained imitation learning policies for cooking-style manipulation. It reports success gains of 15–25 percentage points over RGB-only policies on contact-rich skills, alongside lower operator workload in a user study. The idea is practical, but the experiments do not establish reliability across every tool or industrial environment.

Sources

SwapBot: Tool-Centric Robot Manipulation via Interchangeable End-Effectors and Tactile Sensing · 2026-10-29

Gym-Anything lets coding agents build the robot training worlds

A robot policy can look capable in a familiar simulator and still fail when the world changes. Gym-Anything attacks the supply side of that problem: coding agents create and audit new physical simulations, rather than waiting for experts to hand-build each environment. The resulting RoboWorld benchmark spans 25 simulations and 25 test tasks.

The researchers report an average reward of just 0.06 for the best frontier models, with no test task solved. They also explore repeated attempts, privileged teachers and demonstration-generating code policies. The work broadens the testing ground for embodied AI and sim-to-real research, but a convincing simulation is not automatically a faithful model of factory physics.

Sources

Gym-Anything for Robotics: Scaling physical simulations with digital agents · 2026-10-29

TouchBridge carries human fingertip contact into robot training

When a person turns an object in the hand, the important information is not just where the fingers went. It is where they touched and how the contact changed. TouchBridge combines a wearable demonstration interface with tactile-conditioned retargeting, preserving fingertip contact

relations as examples are converted for a robot hand.

On four real tasks, the resulting VLA policy reached 72.5% average success, versus 77.5% with the same number of directly teleoperated robot demonstrations. Human data collection was 7.0 times faster. That trade-off could be attractive when robot time is scarce, though the gap in success and the narrow evaluation still matter.

Sources

TouchBridge: Tactile-Conditioned Human-to-Robot Transfer for Dexterous Manipulation · 2026-10-29

LOGIV checks whether the world still supports the robot's plan

Completing a step is not the same as preserving its result. A robot may place a part correctly, then knock it aside while executing the next instruction. LOGIV introduces an explicit task graph that records which later actions depend on which physical conditions, checks those conditions against observations and repairs the broken links locally.

The framework sits above existing VLA or world-action policies and adds no trainable component. In three real-robot tasks, successful trials rose from 12 out of 30 to 17 out of 30; broader dual-arm tests also showed gains. The engineering promise is recoverable long-horizon execution, provided the task graph and observations capture what actually matters.

Sources

LOGIV: Graph-Verified Execution with Causal Local Repair for Long-Horizon Robot Tasks · 2026-10-29

Aeon and Mujin put mixed-case palletizing into a live Japanese distribution hub



AI-generated illustration · Conceptual AI-generated illustration of mixed-case palletizing: a suction-equipped arm holds a carton above a pallet-supported stack while an overhead camera observes the placement area. Not a depiction of actual Aeon or Mujin equipment.

Mixed cartons rarely arrive in the neat order that a fixed palletizing script expects. At Aeon's Kawaguchi East XD, launched on October 29 with Mujin, the system simulates a loading plan for each order, then uses 3D vision to check the evolving stack and adjust the robot's movements if boxes shift. That feedback loop is the engineering heart of the deployment.

The WES coordinates robot arms, AGVs, shuttles and worker instructions against store orders and deadlines. Unlike a staged laboratory demonstration, this is a declared operating distribution hub. Yet the published description is from the companies themselves: throughput, uptime and savings need independent, sustained operational evidence.

Sources

イオン物流構造改革の次世代自動化モデルセンター「イオン川口東ＸＤ」稼働 · 2026-10-29

Agility updates the capital available for scaling Digit 5

The distance from a convincing humanoid demonstration to a serviceable fleet is measured partly in capital. Agility Robotics and Churchill XI now expect roughly $465 million in gross proceeds from their proposed combination, according to a filing dated October 30. The companies target closing on November 4 and Nasdaq trading on November 5, subject to approval and other conditions.

Agility says early Digit 5 deliveries are planned for the first half of 2027 and cites more than $300 million in multiyear orders. Those are commitments and plans, not revenue already earned or evidence of a new control breakthrough. The question for Physical AI is how financing translates into uptime, support and repeatable customer work.

Sources

Agility Robotics and Churchill Capital Corp XI Announce Expected Gross Capital Raise of Approximately $465 Million · 2026-10-30

Opentrons raises $40 million to connect AI experiments with physical labs

A language model can propose a thousand experiments before a human laboratory has finished one. Opentrons is investing in the slower half of that loop: liquid-handling robots, software and structured records of what was actually done. On October 29 it announced a $40 million Series E to expand the infrastructure for AI-directed laboratory work.

The company reports more than 11,000 robots deployed across over 80 countries. An open API and repeatable physical execution matter because scientific models need trustworthy outcomes, not just plausible hypotheses. The

release documents a financing plan and installed-base claim, however, not independent proof that an autonomous laboratory can discover reliably without human oversight.

Sources

Opentrons Announces $40M Series E Round to Standardize AI-Driven Science on Its Robots · 2026-10-29

Bleu Robotics bets on few-shot teaching rather than general-purpose hype

Industrial robotics has a stubborn problem: the next product variant can invalidate yesterday's carefully programmed cell. Bleu Robotics, an Inria spinout, has raised €5 million to pursue a narrower alternative to universal robot intelligence. Its few-shot imitation learning approach tries to infer a workstation task from roughly ten operator demonstrations, using existing perception and motion-control tools.

The company's earlier Bleu_0.1 material reports promising controlled experiments and aims for task deployment in under an hour. It also states clear boundaries: deformable materials are outside its present focus, and the robot stops when an unfamiliar trajectory appears. The funding is real news; factory-scale robustness remains a claim to test.

Sources

French startup Bleu Robotics secures €5 million to teach humanoid robots new tasks with fewer demonstrations · 2026-10-29

Introducing Bleu_0.1 · 2026-03-17

Analysis

Physical AI shifts from isolated demonstrations to systems that can be operated

Tool changing, tactile data, verified task dependencies and live warehouse feedback make the surrounding engineering stack as important as the foundation model.

Measure contact success, recovery after disturbances, integration effort and operational uptime alongside policy benchmarks.

Sources

SwapBot: Tool-Centric Robot Manipulation via Interchangeable End-Effectors and Tactile Sensing · 2026-10-29

Gym-Anything for Robotics: Scaling physical simulations with digital agents · 2026-10-29

TouchBridge: Tactile-Conditioned Human-to-Robot Transfer for Dexterous Manipulation · 2026-10-29

LOGIV: Graph-Verified Execution with Causal Local Repair for Long-Horizon Robot Tasks · 2026-10-29

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Introducing Bleu_0.1 · 2026-03-17

Issue glossary

Terms and definitions from the editorial dataset.

VLA

Vision-Language-Action model: turns visual observations and instructions into robot actions.

Tactile sensing

Measuring physical contact or force through sensors on a tool or hand.

Imitation learning

Learning a robot policy from demonstrated actions.

End-effector

A robot's tool or gripper attached at the end of its arm.

Sim-to-real

Transfer of policies or insights from simulation to physical hardware.

Task graph

Representation of actions and the conditions on which later steps depend.

WES

Warehouse Execution System: coordinates equipment and operational work.

3D vision

Visual sensing that estimates object geometry and spatial placement.

Palletizing

Arranging boxes or goods into stable loads for transport.

Few-shot

Adapting to a new task from a small number of examples.

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SwapBot: Tool-Centric Robot Manipulation via Interchangeable End-Effectors and Tactile Sensing · 2026-10-29

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